

MATLAB Lab Viva Questions & Answers

Based on the uploaded IT Workshop / MATLAB Lab Manual — Experiments 1–10

Experiment 1 — Introduction to MATLAB

1. What is MATLAB?

MATLAB is a high-performance programming language and environment used for technical computing, numerical computation, visualization, and programming.

2. What does MATLAB stand for?

MATLAB stands for MATrix LABoratory.

3. Why is MATLAB called Matrix Laboratory?

Because MATLAB was originally developed to provide easy access to matrix computations, and matrices are fundamental to MATLAB.

4. What are the main uses of MATLAB?

Numerical computation, matrix operations, data visualization, mathematical modeling, programming, image processing, and engineering simulations.

5. What is the Command Window?

It is the area where we enter MATLAB commands, variables, functions, and execute M-files.

6. What is the Workspace?

The Workspace contains the variables currently created during a MATLAB session and stored in memory.

7. What is Command History?

It stores commands previously entered in the Command Window so they can be viewed, copied, or executed again.

8. What is Current Directory?

It is the directory MATLAB uses as a reference for file operations.

9. Difference between *clc* and *clear*?

clc clears the Command Window; *clear* removes variables from the Workspace.

10. What does *clear all* do?

It clears all variables from the Workspace.

11. What is the use of *help*?

help topic displays help information about a MATLAB topic or function.

12. What is *lookfor*?

lookfor searches MATLAB help topics using a specified keyword.

13. How do you create a variable?

Using `variable = value`, for example `x = 10`;

14. Does MATLAB require variable declaration?

No. MATLAB variables are dynamically created when a value is assigned.

15. What is the purpose of a semicolon?

It suppresses the display of output in the Command Window.

Experiment 2 — Variables and Miscellaneous Commands

16. Can a MATLAB variable be overwritten?

Yes. Assigning a new value to the same variable replaces its previous value.

17. What is operator precedence?

It determines the order in which mathematical operations are performed.

18. What is $2 + 3 * 4$?

14, because multiplication has higher precedence than addition.

19. What is $(2 + 3) * 4$?

20, because parentheses have higher precedence.

20. What is format short?

It displays floating-point numbers using a shorter format.

21. What is format long?

It displays floating-point numbers with greater precision.

22. What is format bank?

It displays numbers with two digits after the decimal point.

23. Difference between who and whos?

who lists variables; whos gives detailed information such as name, size, bytes, and class.

24. How do you delete only variable p?

Use clear p.

25. How do you enter multiple statements on one line?

Use a comma or semicolon between statements. A comma allows output; a semicolon suppresses it.

Experiment 3 — Matrix Operations

26. What is a matrix?

A two-dimensional array of numbers arranged in rows and columns.

27. How do you create a matrix?

Example: $A = [1 \ 2; \ 3 \ 4];$

28. How are rows separated?

Rows are separated using a semicolon (;).

29. What is a scalar?

A scalar is a 1×1 matrix.

30. What is a row vector?

A vector containing one row, such as $[1 \ 2 \ 3 \ 4]$.

31. What is matrix multiplication?

$A * B$. The number of columns of A must equal the number of rows of B.

32. What is element-wise multiplication?

$A .* B$ multiplies corresponding elements.

33. Difference between * and .*?

* is matrix multiplication; .* is element-by-element multiplication.

34. What is matrix division?

`/` is matrix right division and `./` is element-wise division.

35. What is transpose?

`A'` converts rows into columns and columns into rows.

36. What does `det(A)` do?

It calculates the determinant of `A`.

37. What does `inv(A)` do?

It calculates the inverse of `A` when the inverse exists.

38. What does `rank(A)` calculate?

It calculates the rank of a matrix.

39. Can matrices of different dimensions be added?

Generally no; matrix addition requires compatible dimensions, normally the same size.

40. Is matrix multiplication commutative?

No. In general $A*B$ is not equal to $B*A$.

41. What is a null matrix?

A matrix whose elements are all zero.

42. What is a unity/identity matrix?

A square matrix with 1 on the main diagonal and 0 elsewhere.

Experiment 4 — Built-in Functions & Linear Equations

43. What are built-in functions?

Functions already provided by MATLAB for commonly required mathematical and computational operations.

44. Give examples of scalar functions.

`sin`, `cos`, `tan`, `exp`, `log`, `abs`, `sqrt`, `round`, `floor`, and `ceil`.

45. What does `sqrt()` do?

Calculates the square root.

46. What does `abs()` do?

Returns the absolute value.

47. What does `round()` do?

Rounds a number toward the nearest integer.

48. Difference between `floor()` and `ceil()`?

`floor` rounds toward negative infinity; `ceil` rounds toward positive infinity.

49. What does `max()` do?

Returns the largest element.

50. What does `min()` do?

Returns the smallest element.

51. What does `length()` do?

Returns the length of a vector.

52. What does `sort()` do?

Sorts elements in ascending order by default.

53. What does `sum()` do?

Calculates the sum of elements.

54. What does `mean()` do?

Calculates the arithmetic mean.

55. What does `std()` calculate?

Calculates the standard deviation.

56. What does `size(A)` return?

It returns the dimensions of matrix A.

57. What does `eye()` generate?

An identity matrix.

58. What does `zeros()` generate?

A matrix containing only zeros.

59. What does `ones()` generate?

A matrix containing only ones.

60. What does `diag()` do?

It extracts the diagonal of a matrix or creates a diagonal matrix.

61. What does `eig()` calculate?

Eigenvalues and eigenvectors of a matrix.

62. What does `rref()` mean?

Reduced Row Echelon Form.

63. How can linear equations be solved using matrices?

For $AX = B$, MATLAB can solve the system using $X = A \backslash B$.

Experiment 5 — Plotting Data

64. Why do we use plots?

Plots visually represent data or mathematical functions, making trends and relationships easier to understand.

65. What is the basic plotting command?

`plot(x,y)`

66. What is `fplot()`?

It plots a mathematical function over a specified range.

67. Difference between `plot()` and `fplot()`?

`plot()` generally plots given data vectors; `fplot()` can directly plot a function over an interval.

68. How do you give a title?

`title('My Graph')`

69. How do you label the x-axis?

`xlabel('Time')`

70. How do you label the y-axis?

`ylabel('Amplitude')`

71. What is a legend?

It identifies different datasets or curves in a graph.

72. How do you display grid lines?

`grid on`

73. What does *hold on* do?

It allows multiple plots to be drawn on the same figure.

74. What does *hold off* do?

It returns plotting to normal behavior.

75. What is a line specifier?

It specifies line style, color, and marker.

76. What does *r* mean in a plot specification?

Red.

77. What does *b* mean?

Blue.

78. What does *g* mean?

Green.

79. What does *--* mean?

Dashed line.

80. What does *:* mean?

Dotted line.

81. What does *o* mean?

Circle marker.

82. What does *s* mean?

Square marker.

Experiment 6 — MATLAB M-Files

83. What is an M-file?

A MATLAB file with the `.m` extension containing MATLAB commands or a function.

84. What is a script file?

A sequence of MATLAB commands executed in the order in which they are written.

85. Why use script files?

They allow a program to be saved, edited, corrected, and executed repeatedly.

86. What is a function file?

A MATLAB program that accepts input arguments and returns output arguments.

87. Main difference between script and function?

A script uses the current/base workspace, while a function has its own workspace and can use input/output arguments.

88. What is the syntax of a function?

function output = functionName(input)

89. Can a function have multiple outputs?

Yes. Example: function [area,perimeter] = rectangle(l,w)

90. What are local variables?

Variables defined inside a function and recognized only within that function.

91. What is a global variable?

A variable shared between functions or the workspace using the global command.

Experiment 7 — Debugging M-Files

92. What is debugging?

Systematic identification and correction of errors in a program.

93. What are three common types of errors?

Syntax errors, runtime errors, and logical errors.

94. What is a syntax error?

An error caused by violating MATLAB programming syntax.

95. What is a runtime error?

An error that occurs while the program is executing, such as accessing an invalid array index.

96. What is a logical error?

The program runs but produces an incorrect result because the logic is wrong.

97. What is a breakpoint?

It temporarily stops execution so variables and program behavior can be inspected.

98. What is dbstop?

A command used to set a debugging breakpoint.

99. What is dbquit?

It exits debugging mode.

100. What are F10 and F11 used for?

F10 = step over; F11 = step into. F5 continues execution.

101. What is wrong with `length = 5; area = length * width;`?

width has not been defined.

102. What is wrong with `avg = sum(x)/length(x)^2`?

The average should be `sum(x)/length(x)`.

103. What happens with `a = 10; b = 0; c = a/b;`?

Division by zero occurs; MATLAB may produce Inf for floating-point division.

104. What is wrong with `factorials(n)`?

The MATLAB function is `factorial(n)`.

105. What is wrong with `A=[3 5 7 9]; val=A(5)`?

A has only four elements, so A(5) is out of bounds.

106. What is wrong with $C=25$; $F=(9/5)*c+32$?

MATLAB is case-sensitive; C and c are different variables.

107. What is wrong with $y=x*x+;$?

There is an incomplete expression after +; it is a syntax error.

108. What is wrong with $total=total+x$ in an average loop?

It should add the individual element $x(i)$, not the complete vector.

109. What is wrong with $fib(n-1)+fib(n-3)$?

For the standard recursive Fibonacci relation, the second call should be $fib(n-2)$.

110. What is wrong with $disp(['Name: ' name ' Age: ' age])$?

age is numeric and should be converted using $num2str(age)$.

Experiment 8 — Control Flow & Operators

111. What is control flow?

It determines the order in which program statements are executed.

112. What are relational operators?

$<$, $>$, $<=$, $>=$, $==$, and $\sim=$.

113. Difference between $=$ and $==$?

$=$ is assignment; $==$ is equality comparison.

114. What does $\sim=$ mean?

Not equal to.

115. What are logical operators?

$\&$, $|$, and $\sim$, representing AND, OR, and NOT.

116. What does $\&$ mean?

Logical AND.

117. What does $|$ mean?

Logical OR.

118. What does $\sim$ mean?

Logical NOT.

119. What is an if statement?

It executes a block only when a specified condition is true.

120. What is if-else?

It executes one block if the condition is true and another if it is false.

121. What is if-elseif-else?

It tests multiple conditions sequentially.

122. What is switch-case?

A control structure that selects a block based on the value of an expression.

123. What is a for loop?

It repeatedly executes a block for a specified range or set of values.

124. What is a while loop?

It repeatedly executes statements while a condition remains true.

125. Difference between for and while?

for is generally used when the iteration range is known; while continues until its condition becomes false.

126. What is a nested loop?

A loop placed inside another loop.

127. What does break do?

It terminates the current loop.

128. What does continue do?

It skips the remaining statements in the current iteration and starts the next iteration.

Experiment 9 — Image Acquisition & Pre-processing

129. What is image acquisition?

Capturing or importing an image into MATLAB for processing.

130. How do you read an image?

```
I = imread('image.jpg');
```

131. How do you display an image?

```
imshow(I)
```

132. What is image preprocessing?

Preparing a raw image for analysis by improving quality, reducing noise, and normalizing it.

133. What is RGB?

Red, Green, Blue — three channels commonly used to represent a color image.

134. How do you convert RGB to grayscale?

```
gray = rgb2gray(I);
```

135. Why convert to grayscale?

It reduces three color channels to one intensity channel, simplifying processing.

136. What is HSV?

Hue, Saturation, and Value.

137. How do you convert RGB to HSV?

```
hsvImage = rgb2hsv(I);
```

138. What is image resizing?

Changing image dimensions, e.g. `imresize(I,0.5)` reduces the size by half.

139. What is cropping?

Extracting a specific region of interest from an image.

140. What is normalization?

Adjusting image intensity values to a desired range; the manual uses `mat2gray()`.

141. What is image noise?

Unwanted random variations in pixel intensity.

142. Why is preprocessing important?

It improves reliability of later tasks such as segmentation, feature extraction, and recognition.

Experiment 10 — CNNs & Computer Vision

143. What is CNN?

CNN stands for Convolutional Neural Network, a deep-learning architecture commonly used for image data.

144. What are the main components of a CNN?

Convolutional layers, pooling layers, and fully connected layers.

145. What is a convolutional layer?

It extracts features from an image using learnable filters/kernels.

146. What is a pooling layer?

It downsamples feature maps and reduces spatial dimensions.

147. What is a fully connected layer?

It connects extracted features for final prediction/classification.

148. What is image classification?

Assigning a class/category to an entire image.

149. What is image segmentation?

Assigning a class label to each pixel.

150. What is semantic segmentation?

Assigning a semantic class to every pixel in the image.

151. What is object detection?

Identifying objects and their locations, usually using bounding boxes.

152. Difference between classification and detection?

Classification asks what is in the image; detection asks what and where.

153. Difference between segmentation and detection?

Detection generally uses bounding boxes; segmentation identifies pixels belonging to each class/object.

154. Name CNN models in the manual.

AlexNet, VGG16, and ResNet.

155. What is transfer learning?

Using a pretrained model and adapting it to a new dataset/task.

156. What is YOLO?

YOLO stands for You Only Look Once and is an object-detection approach.

157. What is Faster R-CNN?

A deep-learning-based object detection model using region-based detection.

158. Name segmentation networks in the manual.

U-Net, SegNet, and DeepLab.

159. What is required for object detection training?

Training images together with bounding-box labels.

160. What is the CNN classification workflow?

Load dataset, split training/testing data, load a pretrained CNN, replace the final classification layer, train, and evaluate accuracy.

Top 30 Questions to Memorize First

1. What is MATLAB?
2. What does MATLAB stand for?
3. What is Workspace?
4. Difference between `clc` and `clear`.
5. Difference between `who` and `whos`.
6. What is an M-file?
7. Difference between script and function.
8. What is a matrix?
9. Difference between `*` and `.*`
10. What is transpose?
11. What does `det()` do?
12. What does `inv()` do?
13. What does `rank()` do?
14. What are `zeros()`, `ones()` and `eye()`?
15. What is `size()`?
16. What is `plot()`?
17. What is `fplot()`?
18. What is a legend?
19. What are line specifiers?
20. Difference between `=` and `==`.
21. What are relational operators?
22. What are logical operators?
23. Difference between `for` and `while`.
24. Difference between `break` and `continue`.
25. What is debugging?
26. What is a breakpoint?
27. What is image preprocessing?
28. Difference between RGB and grayscale.
29. What is CNN?
30. Difference between classification, segmentation and object detection.

Important command cheat sheet

Command	Meaning
<code>clc</code>	Clear Command Window
<code>clear</code>	Remove variables
<code>clear all</code>	Remove all variables
<code>who</code>	List variables

Command	Meaning
whos	Detailed variable information
help	Display help
lookfor	Search help by keyword
format short	Short numerical display
format long	Long numerical display
format bank	Two decimal places
size(A)	Size of matrix
length(A)	Length
det(A)	Determinant
inv(A)	Inverse
rank(A)	Rank
rref(A)	Reduced Row Echelon Form
eig(A)	Eigenvalues/eigenvectors
zeros()	Zero matrix
ones()	Matrix of ones
eye()	Identity matrix
plot()	2-D plotting
fplot()	Function plotting
title()	Plot title
xlabel()	X-axis label
ylabel()	Y-axis label
legend()	Plot legend
grid on	Grid
imread()	Read image
imshow()	Display image
rgb2gray()	RGB to grayscale
rgb2hsv()	RGB to HSV
imresize()	Resize image
mat2gray()	Normalize image
dbstop	Set breakpoint
dbquit	Exit debugging

Note: The uploaded manual is labeled IT Workshop / Scilab Lab in the course information, but the experiments themselves are written primarily for MATLAB. This PDF follows the MATLAB terminology and syntax used in those experiments.